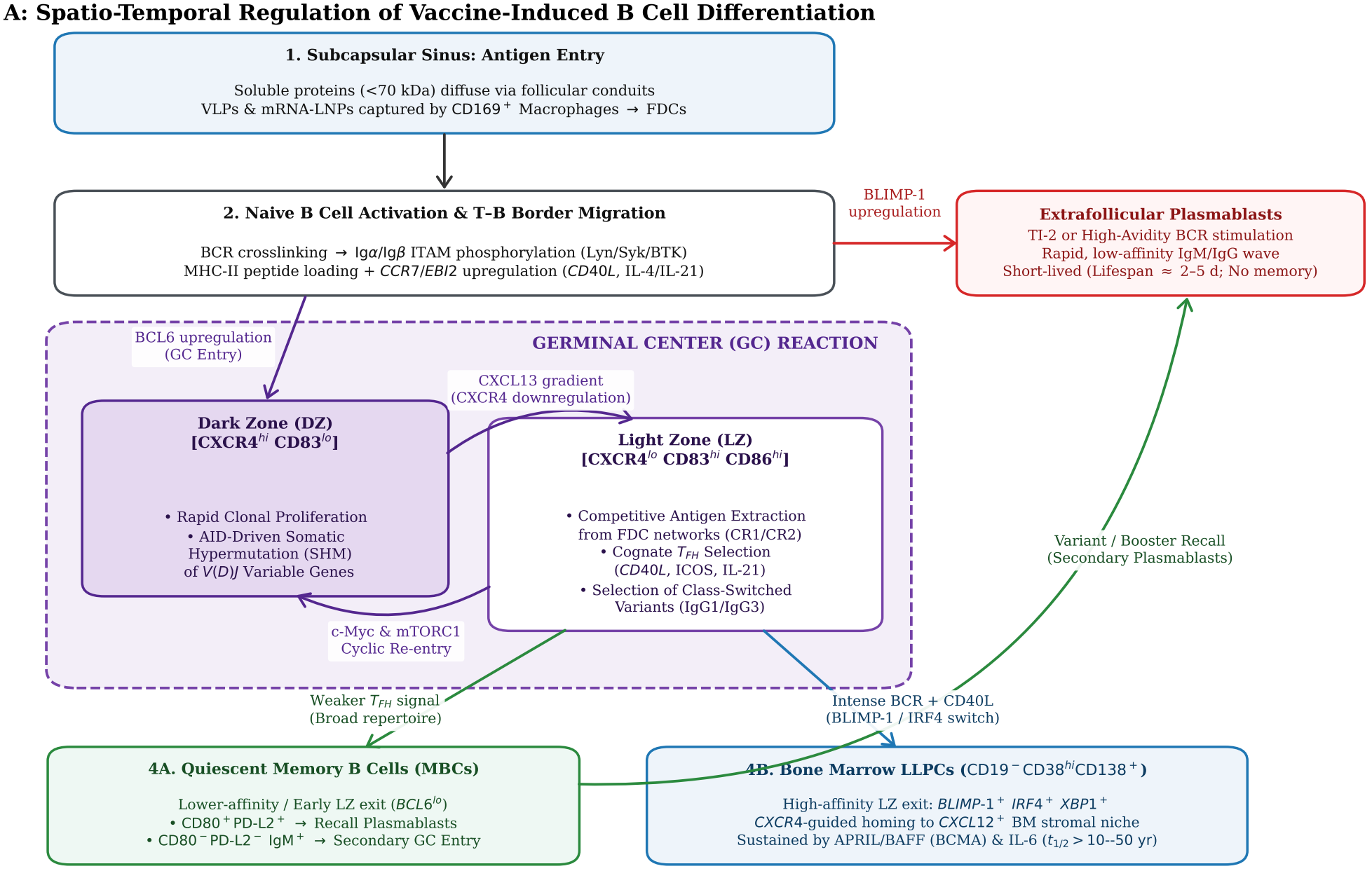

A: Spatio-Temporal Regulation of Vaccine-Induced B Cell Differentiation
1. Subcapsular Sinus: Antigen Entry
Soluble proteins (<70 kDa) diffuse via follicular conduits
VLPs & mRNA-LNPs captured by CD169+ Macrophages → FDCs
2. Naive B Cell Activation & T–B Border Migration
BCR crosslinking → Igα/Igβ ITAM phosphorylation (Lyn/Syk/BTK)
MHC-II peptide loading + CCR7/EBI2 upregulation (CD40L, IL-4/IL-21)
BLIMP-1 upregulation
Extrafollicular Plasmablasts
TI-2 or High-Avidity BCR stimulation
Rapid, low-affinity IgM/IgG wave
Short-lived (Lifespan ≈ 2–5 d; No memory)
BCL6 upregulation (GC Entry)
GERMINAL CENTER (GC) REACTION
CXCL13 gradient (CXCR4 downregulation)
Dark Zone (DZ) [CXCR4hi CD83lo]
• Rapid Clonal Proliferation
• AID-Driven Somatic Hypermutation (SHM) of V(D)J Variable Genes
Light Zone (LZ) [CXCR4lo CD83hi CD86hi]
• Competitive Antigen Extraction from FDC networks (CR1/CR2)
• Cognate TFH Selection (CD40L, ICOS, IL-21)
• Selection of Class-Switched Variants (IgG1/IgG3)
c-Myc & mTORC1 Cyclic Re-entry
Variant / Booster Recall (Secondary Plasmablasts)
Weaker TFH signal (Broad repertoire)
Intense BCR + CD40L (BLIMP-1 / IRF4 switch)
4A. Quiescent Memory B Cells (MBCs)
Lower-affinity / Early LZ exit (BCL6lo)
• CD80+PD-L2+ → Recall Plasmablasts
• CD80−PD-L2− IgM+ → Secondary GC Entry
4B. Bone Marrow LLPCs (CD19−CD38hiCD138+)
High-affinity LZ exit: BLIMP-1+ IRF4+ XBP1+
CXCR4-guided homing to CXCL12+ BM stromal niche
Sustained by APRIL/BAFF (BCMA) & IL-6 (t1/2 > 10--50 yr)